

Development and temporal evaluation of prehospital prediction models for nine emergency outcomes: overall and specialty-specific EMT cohorts

Research manuscript update, 7 September 2026. This version integrates completed specialty P1_04 results, all 324 matched current-model intervals, 270 prespecified paired contrasts, both specialty descriptive cohorts, all 27 required best-tree case panels and a real aggregate-analysis reproduction package. Authors' declarations and source-specific clinical judgments are itemized separately. Scientific/formal acceptance of this report remains false.

Abstract

Objective. To develop and temporally evaluate models for nine emergency outcomes using information available before hospital arrival, and to examine independently developed medicine and surgery extensions.

Design and setting. Single-centre retrospective linked emergency medical service and hospital records from Taipei Tzu Chi Hospital, with Taipei and New Taipei EMS sources. Development used 2020-2023, temporal validation used 2024 and frozen evaluation used 2025. The existing methods document reports 24,784 development encounters/19,944 patient groups, 7,504 validation encounters/6,600 groups and 9,504 frozen encounters/8,020 groups; final model-specific denominators are retained separately. [Source: method_TW.md; source_id method.]

Methods. Patient-disjoint folds, fold-local preprocessing, development-based model and calibration procedures, locked 2024 operating points and immutable model/spec identities governed evaluation. Twelve final model families span clinical scoring, regression, gradient boosting, neural/text models, foundation models and ensembles. OP_NAME-defined medicine/surgery branches were additional independently trained analyses. Frozen results were not used for reselection.

Results. The accepted overall rectangle covers 108 outcome/family tasks; specialty research covers 216. Frozen specialty evaluation and its additive delivery passed independent validation. The integrated primary registry has 324 displayed model/outcome/group records and 324 primary direct explanations; it is unchanged by this report. Current validation intervals cover all 324 models, and the 270 prespecified paired contrasts and both specialty descriptive cohorts are complete. A newly completed locked-model P3 2024 sensitivity yielded AUROC 0.922453 overall, 0.921886 after excluding recorded field ETT and 0.918131 after excluding recorded ETT/SGA. Full estimates, denominators and intervals accompany this manuscript. The completed specialty robustness program covers 494 task dispositions. Vital-only AUPRC is descriptively lower than the full-input comparator in all 18 specialty/outcome combinations; corrected time-removal effects vary by outcome. Nine Surgery full-complete-case analyses are not estimable. [Sources: overall_accepted; specialty_research; specialty_fv0; direct324; p3_successor; p3_metrics.]

Conclusions. The existing results support description of temporal predictive performance under the locked contracts. They do not establish clinical effectiveness, fairness across all relevant groups or transportability to another hospital. The completed sensitivity program identifies input and population dependence, with rare-event and complete-case limitations. Prospective clinical benefit is not established.

Introduction

Prehospital emergency records contain physiological measurements, clinical assessments and treatments collected before hospital arrival. This study evaluates whether such information supports prediction of mortality, airway-related outcomes, admission and prolonged stay. Its comparison spans clinical scores and several model families within a common temporal framework. Predictions could inform further research on early risk assessment; this study does not evaluate whether acting on model outputs improves patient outcomes.

The specialty extension addresses a distinct research question: how independently developed models behave in OP_NAME-defined medicine and surgery cohorts. It is not an audit that applies one overall winner to two subgroups. Because the specialty extension was added after overall frozen evaluation, the chronology is explicitly disclosed. A prospective Spanish ALS cohort studied NEWS2 for early in-hospital mortality, whereas an unselected Danish ambulance cohort found only moderate performance for commonly used early-warning scores. These differing populations and time horizons motivate the present comparisons but do not provide a directly comparable AUPRC target. Martín-Rodríguez et al., 2020 (<https://pubmed.ncbi.nlm.nih.gov/32169257/>); Lindskou et al., 2023 (<https://pmc.ncbi.nlm.nih.gov/articles/PMC10413164/>).

Methods

Data, linkage and participants

The existing methods document describes deterministic linkage through the rescue identifier recorded by EMS and hospital triage. Identifiers were used for linkage and were excluded as predictors. Eligibility required an EMT record and valid age ≥ 18 . Records were deduplicated under the fixed source-completeness rule. Hash-based patient groups governed temporal isolation and resampling. The source dataset and successive cohort exclusions are documented in method_TW.md; the current report includes the Overall baseline/missingness/flow aggregates with their original lineage and newly completed descriptive tables for both exact specialty populations. The source-bound flow appendix now follows 34,428 eligible through-2024 records, removal of 179 duplicates and removal of 1,961 development records for patient isolation, leaving 24,784 development and 7,504 validation records. The current 7,289-record Overall evaluation population is separately labelled; its difference from the native specialty populations is not reconstructed as an unsupported exclusion identity. This manuscript has not independently audited original IRB paperwork or raw linkage quality. [Source: method.]

The methods document records institutional approval 15-IRB101 and waiver of informed consent for retrospective deidentified analysis. Authors should confirm the recorded institution, approval number and waiver against the source approval letter before submission. [Source: method, ethics section.]

Medicine includes exact 急診内科 OP_NAME and surgery exact 急診外科 under the accepted normalization contract. Other/missing values remain only in overall. OP_NAME is a hospital-defined router; it is not a predictor or outcome. The time/source limitation and owner-approved cohort definition must remain visible in interpretation. Group-specific and model-specific denominators must not be pooled into a single common N without proof of row identity. [Sources: method_qa; specialty_research.]

Outcomes and prediction time

The nine binary outcomes are P1 emergency-department mortality, P2 hospital mortality, P3 the recorded ENDO-related intubation outcome, S1/S2 mortality within three/seven days, S3 specified observation/ward admission dispositions, S4 ICU admission, S5 ICU stay ≥ 14 days and S6 total hospital stay ≥ 30 days. Mortality-equivalent critical discharge remains in P1/P2 under the current method. Missing ENDO maps to negative under the locked definition; missing hospital days for S1/S2 derivation maps to zero. The exact source codes, time horizons and missing-outcome rules are in the bound methods/specs and must accompany submission; broad explorer labels are not operational definitions. [Sources: method; overall_accepted.]

The index is before hospital arrival. Arrival vitals and unapproved hospital-derived variables are excluded from the primary predictor allowlist. Authorized prehospital interventions, including airway care, remain legitimate recorded predictors under the study definition. Their inclusion does not establish causal direction or certainty of the outcome. +ED-early and congestion analyses are separately labelled comparisons. [Sources: method_qa; stage1a_manifest; effective-method addendum.]

Development, calibration and model selection

Preprocessing and base-model fitting use development data. Median imputation, missingness indicators, scaling and derived-score recomputation are fitted within the relevant training fold. Patient-disjoint, outcome-specific stratified group folds use locked seeds and assignments. The accepted model registry specifies each family's exact variant and hyperparameters. This document does not replace those locks with a generic model list. Raw/Platt/isotonic calibration follows the accepted development OOF cross-fitting method. [Sources: method; method_qa; overall_accepted; specialty_research.]

The final comparison has twelve families per outcome. TabPFN uses the accepted outcome-specific variants; a missing historical standard-cloud S1-S6 task is not a gap. Some specialty meta-models use a separate validation-only meta contract and must be labelled accordingly. Surgery/P1 has no valid single cross-family representative under its current contract. The principal clinical score rSI-sMS uses a native complete-component cohort and is not directly ranked against full-population models. Earlier score and conventional-baseline results are included as their distinct development/validation supplement. [Sources: specialty_report; effective-method addendum.]

Validation and statistical analysis

Threshold selection uses 2024: first seek sensitivity/specificity both ≥ 0.90 ; otherwise maximize their minimum, then Youden index and then the higher threshold. The overall stacking outcomes required the fallback. Threshold-dependent 2024 metrics are operating-point selection estimates; 2025 evaluates the unchanged threshold. Discrimination, calibration, Brier, PPV/NPV, F1, confusion matrices and DCA are reported under their locked contracts. DCA compares model/treat-all/treat-none over 0.01-0.15 by 0.005. [Sources: method; method_qa.]

The current Stage2a-derived uncertainty analyses use 2,000 outcome-stratified disjoint patient-group bootstrap replicates and percentile intervals, with estimability conditions. The older accepted Overall W02 ablation intervals instead use outcome-stratified row resampling in their exact bound runner; their row-level unit and generation are retained explicitly, not relabelled as patient-group uncertainty. Paired AUROC uses DeLong and paired AUPRC uses cluster bootstrap under exact common-row alignment. Nominal P values are not multiplicity-adjusted. All 324 current-model validation tasks now have matched intervals, and all 270 source-prespecified five-family contrasts are complete on exact common rows. This does not imply an exhaustive all-family comparison or paired intervals for the separate P104 arm differences. [Source: method; requirements R09.]

The frozen stage follows prior locks and one-time authority; historical overall 2025 exposure is disclosed and the holdout is not described as pristine. Specialty chronology and historical specialty exposure are separately recorded. A technical reproduction can be documented only with exact input/code/model/threshold/output identity, without new selection opportunities. [Sources: overall_accepted; specialty_fv0.]

Sensitivity analyses and model explanation

The accepted overall P1_04 and DNR program are reused. They include the predefined ablations and domain/age/CV/congestion comparisons, with formal not-estimable dispositions retained. Current outcome amendments govern interpretation of historical mortality-definition analyses. Full independent specialty P1_04, DNR and missingness results are now incorporated from their separately validated 494-task packet, with limited and not-estimable components retained. [Sources: overall_p104; overall_dnr; effective-method addendum.]

The new overall P3 comparison uses the originally locked LightGBM with development-OOF Platt calibration and fixed threshold, reusing saved 2024 probabilities. It separately evaluates the full cohort, no recorded field ETT and no recorded ETT/SGA. There were no new model fits, predictions, threshold selections or raw2025 reads. [Source: p3_successor.]

The final primary explanations are direct outputs of the hash-bound models, with 648 separate beeswarm/bar assets. Exact algorithms, target outputs, background/explanation samples and approximation diagnostics remain in task receipts. A direct SHAP explanation is not a causal effect, and an ensemble-member contribution is not automatically attribution to an original patient feature. Archived surrogate explanations remain secondary audits. The source-prespecified best-tree TP/FN and age-GCS coverage is complete for all 27 group/outcome cells; optional all-model case coverage and population explanation stability are distinct from this achieved requirement. [Sources: direct324; requirements R21-R22.]

Results

Existing primary model results

The current source-bound public metric appendix public_metric_snapshot.csv contains 648 group/outcome/family/split records ($324 \times \text{two splits}$) with the exact displayed values and family-specific N/events. It is a transparent copy of aggregate display data, not newly recomputed patient-level performance. The overall accepted source and specialty independent-validation receipts supply the underlying evidence. Full selected-calibration specialty results and research-locked representative choices remain in the bound specialty report; nothing here chooses a new representative from 2025. [Sources: final_site; overall_accepted; specialty_delivery; specialty_report.]

For illustration of the already locked primary specialty choices, medicine P1/P2 use TabICL and P3 stacking, with frozen AUROC 0.919312/0.869263/0.884958 and AUPRC 0.356768/0.470066/0.388948. Surgery P2 uses TabICL and P3 TabPFN; surgery P1 remains without a valid single representative. These values are copied from the exact research-locked representative table, not evidence of superiority on a common cohort. [Source: specialty_report.]

P3 performance after recorded airway exclusions

2024 population	N	P3 events	AUROC	AUPRC
Allcomer	7,504	285	0.922453	0.417436
No recorded field ETT	7,492	279	0.921886	0.420773
No recorded prehospital ETT/SGA	7,439	253	0.918131	0.408513

The locked model's point estimates change modestly across these evaluation subsets. This is a descriptive nested-cohort comparison, not an independent test proving equivalence. The full 95% intervals, calibration statistics and curves are in `p3_metrics.csv` and the bound source directory; all 2,000 bootstrap replicates succeeded for the reported metrics. [Sources: `p3_metrics`; `p3_successor`.]

Twelve validation records have the ETT flag, but only six are P3-positive. The recorded data therefore do not support the old prompt's assertion that field ETT necessarily implies ENDO=Y. This discrepancy must be reported without relabelling outcomes or inventing its clinical cause. Lack of a recorded field-airway flag does not prove that the remaining positive events began only after arrival. [Source: `p3_flow`.]

Specialty robustness: feature reduction and missingness change the interpretation

The Medicine and Surgery program now has a completed disposition for each of 494 required logical tasks (247 per specialty): 473 fully computed, 12 with a not-estimable component and nine not estimable. All nine fully non-estimable tasks are Surgery full-complete-case analyses; the recorded reason is an empty evaluation or single-class training cohort. This is a failure of that analysis population to support estimation, not zero predictive value. The corrected time-removal definition is present for all 18 specialty/outcome combinations. [Source: `specialty_p104`.]

On the recorded 2024 comparator cohorts, vital-only inputs have lower AUPRC point estimates than full inputs for every one of the nine outcomes in each specialty. Comparison-minus-reference AUPRC ranges from -0.255378 to -0.019432 for Medicine and -0.186535 to -0.024746 for Surgery. This consistent descriptive direction supports reporting that the richer input representation carries predictive information in these fits; individual-arm confidence intervals do not establish a paired difference or causal benefit. The exact arms, N/events and 95% intervals for P1-P3 are reproduced in the quantitative appendix. [Source: `specialty_p104`; `descriptive_point_comparisons.csv`.]

The corrected removal of all planned time intervals has mixed AUPRC changes: -0.008645 to $+0.011651$ for Medicine and -0.010146 to $+0.043590$ for Surgery. Four of nine outcomes in each specialty have negative changes. The aggregate evidence therefore does not support a uniform claim that time features help or harm. Treatment, shock-index, collaborator-variable and distance ablations likewise vary by outcome. The appendix retains all twelve ablations instead of selecting only favorable outcomes. [Source: `specialty_p104`.]

Age, city, DNR and airway exclusions deliberately change the eligible cohort in some arms. Of the 696 existing point-comparison records, 121 are different-cohort comparisons and receive no improvement delta; 575 have exact same-cohort identity and retain descriptive differences. City and rare-event intervals can be broad, so different prevalence and denominators must be considered before comparing discrimination. DNR-blind models, exclusion of DNR confirmed before prediction, and broader DNR exclusions answer different questions; the last can incorporate later documentation and remains a labelled sensitivity. [Source: `specialty_p104`.]

The `P1_04` tables mostly concern recorded raw probabilities and a fixed 0.5 comparator threshold; the threshold-shift and separately calibrated presentations retain their own recorded rules. A negative net benefit at such a threshold is not automatically the net benefit of the final model at its selected clinical operating point. Individual-arm AUROC, AUPRC and Brier intervals, confusion counts, net benefit and

estimability are supplied with each result. Calibrated values are kept with their own sources and intervals. No new threshold, model or calibration selection is made in this report. [Source: specialty_p104.]

Missingness analyses are fully dispositioned, including median/indicator, KNN and iterative approaches and reduced complete-case populations. The nine Surgery full-complete-case results remain not estimable. The twelve limited tasks retain their component-level limitations, especially sparse outcome/subgroup strata; the task-level total must not be read as 494 successful numeric experiments. [Source: specialty_p104; completion_matrix.csv.]

Current uncertainty and descriptive evidence

All 324 current display models now have exact-current 2024 decision curves (28,188 model/treat-all/treat-none rows at the original 1-15% thresholds) as well as matched 2024 confidence intervals for six metrics (1,944 rows), retaining the 30 already-matched tasks and adding only the 294 missing tasks. Their matched calibration curves contain 7,749 rows. The supplement also retains 28,188 stored frozen decision-curve rows in a distinct scope. Reporting those previously computed frozen aggregates did not create a new frozen evaluation or change model selection. The 108 Overall frozen calibration-bin curves remain an explicit absent-source-aggregate gap; available intercept/slope summaries do not reconstruct those bins. This is not a statistical not-estimable claim, and 2024 or specialty curves are not substituted. [Sources: current_supplement_terminal; current_statistics_independent.]

The prespecified five-family comparison set-rSI-SMS, logistic regression, CatBoost, XGBoost and LightGBM-now has 270 paired contrasts, ten per group/outcome. Exact common rows, labels and patient groups were aligned before applying the original paired bootstrap and DeLong methods. Every contrast is estimable. These are nominal unadjusted analyses, not an exhaustive all-family ranking. For Medicine/P1, CatBoost minus XGBoost AUPRC is 0.051562 (paired 95% CI -0.014275 to 0.106023): the point difference alone does not establish a reliable advantage. The full current-model estimates and prespecified comparisons accompany this report. [Sources: current_supplement_terminal; CURRENT_STATISTICS_APPENDIX.md.]

Both specialty descriptive cohorts are complete: Medicine has 15,995 development and 4,770 validation encounters; Surgery has 8,731 and 2,712, respectively. Their 82,728 Table1 rows, 4,536 correlation rows, missingness by variable/year/city and 54 outcome/cohort summaries use the exact locked OP_NAME-derived populations. The earlier Overall descriptive and complete-case tables keep their original cohort labels rather than being substituted for specialty tables. [Source: current_supplement_terminal.]

Baseline context and the remaining text comparison

The earlier strict-v2 Overall reference panel includes 18 traditional score labels and 28 recorded model/score names with calibration, DCA and 90 paired comparisons. It remains a historical reference layer. A separate current-row baseline assembly now contains 675 records, including 27 completed KNN comparators trained independently within their group using the recorded fixed Stage1b defaults and original folds. Existing Overall-trained transferred models are labelled as such, with exact available-subset denominators and same-row CatBoost references; they are not described as specialty-trained models. The baseline interval extension is now complete for all 675 records (2,025 AUROC/AUPRC/Brier intervals), using 630 new unique bootstrap jobs, 18 exact previously computed matches and 27 exact alias reuses. Each interval retains its exact available-row population; it is not a paired improvement estimate. The primary 324-model registry is unchanged. [Sources: current_baseline_points; current_baseline_validation; current_baseline_independent; current_baseline_ci_independent.]

All 36 Medicine/Surgery no-text comparator tasks are complete with 216 independently validated contexts. The additive CPU uncertainty correction is also complete for these pairs and the 18 earlier Overall pairs: 54 pairs, 108 split-specific contrasts and 324 metric/interval rows. Each patient belongs to one outcome stratum defined by its maximum outcome within the split; both probability vectors use shared patient resampling. This corrects the legacy double-weighting of mixed-outcome patients without changing a model, prediction, fold, seed or point estimate. The superseded overlapping-strata intervals remain audit-only. [Sources: `corrected_text_gain`; `corrected_text_export`.]

Across the 36 current specialty 2024 contrasts, 7 nominal paired AUPRC intervals are entirely above zero and 1 entirely below zero. These are descriptive, unadjusted comparisons, not a new model-selection rule. Complete model/outcome values and OOF results are given in `CORRECTED_TEXT_GAIN_APPENDIX.md` and the corrected CSVs. The earlier Overall w03b cohorts remain separately labelled; their results are not substituted for current Overall model performance. Text removal plus refitting under the source-selected settings estimates a predictive comparison, not a causal text effect or prospective clinical benefit.

Required local explanations and their limits

All 27 original best-tree group/outcome requirements now have both a true-positive and a false-negative waterfall (54 required cases), plus 27 age-GCS colored dependence figures. The source-selected trees and raw-model operating points are preserved. Thirty-six additional 2024 case explanations across 21 tree cells were needed to complete those examples; source probabilities matched exactly. The broader supplemental index has 323 dependence figures and 398 waterfalls. Overall/S5 BERT has no varying numeric input in its explanation sample, so a numeric clinical dependence panel is not asserted for that cell. [Sources: `current_shap_best_tree_coverage`; `current_shap_validation`; `current_shap_additive_validation`.]

The original 324-cell explanation sample remains 315 tasks with 20 cases and nine tasks with five. Direct explanation, additive agreement and representative case coverage do not prove stable population feature rankings. BERT whole-text/missingness attribution is not token-level attribution; stacking contributions concern base-model probabilities. The age-GCS view is colored dependence, not SHAP interaction values or a causal estimate. Additional TP/FN coverage for every model is optional and is not silently added as a new GPU requirement.

Overall sensitivities and decision thresholds retain their original scope

The accepted Overall ablation/sensitivity program is now included numerically with its actual outcome-stratified row-bootstrap interval method: 14 source-bound result tables with 550 arm rows, alongside its historical fifteenth not-estimable task and later reported P3 successor. Its nine-outcome DNR program adds 2,862 comparison rows and 162 cohort-flow rows. For Overall/P1, the source 2024 full-input arm has AUPRC 0.471609 (95% CI 0.384276-0.570010), compared with 0.212551 (0.155579-0.287996) for the vital-only arm, both N=7,504/events=107. These are the original research arms, not the different current Overall display cohort. Age, city, CV, mortality, congestion and DNR populations retain their own definitions and chronology. [Sources: `OVERALL_SENSITIVITY_APPENDIX.md`; `overall_sensitivity_source_inventory.json`.]

Recorded net benefit must be interpreted with its threshold. For Medicine/P1, the specialty full/vital-only source arms at threshold 0.5 have net benefits 0.001887 and 0.000419; for Medicine/P2, both recorded values are negative (-0.003145 and -0.018449). These are neither final-model operating-point effects nor causal treatment effects. Decision-curve analysis weights false positives according to the odds of the decision threshold, so discrimination alone does not determine clinical utility. Vickers and Elkin, 2006 (<https://pubmed.ncbi.nlm.nih.gov/17099194/>); [Source: `RECORDED_THRESHOLD_NET_BENEFIT.md`.]

Locked representatives are not newly chosen clinical deployment winners

The operating-point appendix now includes all 324 components and the exact 17 selected specialty research representatives; Surgery/P1 retains no valid single representative. Their selection bases are copied as recorded-development OOF, 2024 validation or 2024 cross-fitting under the applicable specialty contract-and their frozen sensitivity/specificity are reused from stored aggregate tables. For example, Medicine/P1 TabICL uses the locked 0.013448 threshold, with recorded frozen sensitivity 0.843373 and specificity 0.877943. The recorded choice is not changed using those frozen metrics. [Source: LOCKED_RECOMMENDATIONS_AND_OPERATING_POINTS.md.]

The accepted Overall authority fixes 12 components per outcome but contains no separately recorded cross-family deployment recommendation. All 108 Overall operating points are included; no new winner is selected to fill that documentary absence. The dashboard's within-family representative/2024 ranking is distinct from these source-recorded research choices and is not a substitute for a clinical deployment authorization. Authors should confirm any preexisting documented recommendation and intended use, while preserving the separate native rSI-sMS comparator population and uncertainty. These are research candidates, not a demonstrated improvement in clinical care.

Discussion

The study's completed primary evidence supports describing predictive performance within the studied setting and locked temporal design. It does not demonstrate improved care, safety of treatment decisions or broad external validity. Outcome prevalence and model-specific complete-component exclusions affect interpretation, particularly for rSI-sMS. The revised final324/v1 comparison documents changed display scope and differing cohorts; its deltas cannot establish that newer models improved because all 162 shared-family 2024 denominators differ. [Source: final324_vs_v1_comparison.json.]

Surgery/P1 had very few development events, and its lack of a valid representative should remain visible. The current specialty report records nine surgery versus 129 medicine development P1 events. Calibration instability, sample-size adequacy and uncertainty require outcome-specific appraisal; a generic event-per-variable threshold is not evidence of sufficiency for every neural or ensemble model. [Source: specialty_report.]

Further limitations include retrospective single-centre data, city/source differences, missingness, measurement and outcome ascertainment error, mortality-equivalent critical discharge, historical 2025 exposure and the additional-analysis chronology. The specialty router's hospital provenance constrains what “prehospital specialty prediction” can mean operationally. The completed specialty sensitivity results above strengthen the need for outcome-specific interpretation; they do not remove those sources of bias.

Availability, author declarations and submission readiness

The public integrated explorer is recorded at <https://brianann2339.github.io/emt-p1-model-explorer-v3/>. The saved publish receipt supports its state on 1 September 2026; this document does not assert a fresh live-site check. The report/audit ZIP contains aggregate evidence and documentation. The separate portable ZIP reproduces the real aggregate analysis, including all 494 specialty dispositions, 1,659 specialty arm rows, 550 accepted Overall arm rows, the Overall DNR tables, current uncertainty and cohort/operating-point appendices. It checks numerical arithmetic and deterministic outputs after fresh extraction. This is numerical reproduction of aggregate analysis, not retraining or reproduction of bootstrap samples and individual model predictions. The ZIP now also contains seven original scientific entry points with byte-exact local dependencies, separate runtime locks and an executable protected-input planning/dispatch route. Source verification and read-only route tests are executed, but protected model training and heterogeneous GPU runtime reproduction are not. No patient records/models are included, and the route does not dispatch another 2025 unsealing.

Before submission, authors must complete funding, conflicts, registration/protocol availability, patient/public involvement, data-access conditions, author contributions, final clinical-use interpretation, clinical source-measurement/ascertainment judgments and the named remaining model-specific appendices. Unknown declarations must remain unknown until confirmed; absence from this workspace is not a statement that no funding, conflict or involvement exists.

The reporting scaffold follows the TRIPOD+AI framework (<https://www.tripod-statement.org/scope/>) and uses a separate PROBAST+AI appraisal (<https://www.bmj.com/content/388/bmj-2024-082505>). The local reporting map and appraisal now resolve computational evidence and identify specific author-only declarations and source-limited judgments. They are self-assessment evidence maps, not certification of compliance or low risk of bias.

Associated appendices

- P104_QUANTITATIVE_APPENDIX.md: all-analysis effect summaries and primary-outcome intervals.
- CURRENT_STATISTICS_APPENDIX.md: matched current-model intervals and prespecified exact-common-row comparisons.
- EVENTS_AND_FEATURE_DIMENSIONS.md: outcome-specific event/input accounting.
- TRIPOD_AI_REPORTING_MAP.md and PROBAST_AI_APPRAISAL.md: current computational evidence and explicit remaining judgments.
- REQUIREMENTS_CROSSWALK.md: every prior requirement and its latest disposition.
- V1_V3_COMPARISON.md: current source/representation changes without comparing unlike cohorts as model improvement.
- AUTHOR_DECLARATIONS.md: the exact statements only authors can supply.
- portable_analysis.zip: real aggregate replay; original protected model training is outside its executed scope.

Additional source-bound appendices: COHORT_FLOW_APPENDIX.md; OVERALL_SENSITIVITY_APPENDIX.md; LOCKED_RECOMMENDATIONS_AND_OPERATING_POINTS.md; RECORDED_THRESHOLD_NET_BENEFIT.md; LITERATURE_CONTEXT.md. The targeted literature context is not a systematic review.

CORRECTED_TEXT_GAIN_APPENDIX.md documents the completed disjoint-patient correction and supersedes legacy text-gain confidence intervals.

Appendix: current-model temporal uncertainty

All 324 locked display models have matched 2024 intervals. These fixed CatBoost examples illustrate the primary outcomes without selecting a new winner. Intervals are individual-model estimates; the separate 270 paired comparisons use exact common rows and nominal unadjusted inference. Full tables and pair estimates accompany the portable analysis archive.

Group/outcome	N/events	AUROC [95% CI]	AUPRC [95% CI]
overall/P1	7289/60	0.931115 [0.906282, 0.953999]	0.213703 [0.135555, 0.331876]
overall/P2	7289/526	0.908038 [0.896519, 0.918860]	0.489512 [0.448606, 0.531855]
overall/P3	7289/237	0.913556 [0.894931, 0.930884]	0.383623 [0.324951, 0.446564]
medicine/P1	4770/98	0.940944 [0.919655, 0.959322]	0.467949 [0.368685, 0.570886]
medicine/P2	4770/568	0.884963 [0.871309, 0.897859]	0.576214 [0.537891, 0.615329]
medicine/P3	4770/253	0.897502 [0.879326, 0.914477]	0.375752 [0.323158, 0.436678]
surgery/P1	2712/9	0.941135 [0.824151, 0.999505]	0.621538 [0.367724, 0.880362]
surgery/P2	2712/41	0.927478 [0.891331, 0.958619]	0.469710 [0.322491, 0.630076]
surgery/P3	2712/32	0.937687 [0.873237, 0.982661]	0.619920 [0.450514, 0.793709]

For Medicine/P1, the source-prespecified CatBoost minus XGBoost AUPRC difference is 0.051562 (paired 95% CI -0.014275 to 0.106023). This illustrates why model point differences should not be interpreted as a reliable advantage without their paired uncertainty. Model-specific and pair-specific denominators must not be interchanged.

Appendix: events and locked input dimensions

Events per input dimension is descriptive accounting, not effective model degrees of freedom or proof of sample-size adequacy. Overall is the Stage1a candidate input; specialty counts use the P104 logistic comparator input. Surgery/P1 has only nine development events.

Group	Outcome	Development N	Events	Input dimensions	Events/dimension
medicine	P1	15995	129	764	0.1688
medicine	P2	15995	1581	764	2.0694
medicine	P3	15995	703	764	0.9202
medicine	S1	15995	427	764	0.5589
medicine	S2	15995	696	764	0.9110
medicine	S3	15995	6010	764	7.8665
medicine	S4	15995	2086	764	2.7304
medicine	S5	15995	351	764	0.4594
medicine	S6	15995	1018	764	1.3325
surgery	P1	8731	9	764	0.0118
surgery	P2	8731	120	764	0.1571
surgery	P3	8731	56	764	0.0733
surgery	S1	8731	33	764	0.0432
surgery	S2	8731	52	764	0.0681
surgery	S3	8731	1613	764	2.1113
surgery	S4	8731	425	764	0.5563
surgery	S5	8731	34	764	0.0445
surgery	S6	8731	103	764	0.1348
overall	P1	24784	138	766	0.1802
overall	P2	24784	1701	766	2.2206
overall	P3	24784	759	766	0.9909
overall	S1	24784	460	766	0.6005
overall	S2	24784	748	766	0.9765
overall	S3	24784	7664	766	10.0052
overall	S4	24784	2512	766	3.2794
overall	S5	24784	385	766	0.5026
overall	S6	24784	1121	766	1.4634

Appendix: full versus vital-only inputs

2024 research comparators. Individual-arm 95% intervals are retained; no paired interval is inferred. All 18 specialty/outcome vital-only AUPRC point differences are negative. The complete twelve-ablation/sensitivity tables accompany the report in the audit and portable analysis archives.

Group/outcome	Arm	N/events	AUROC [95% CI]	AUPRC [95% CI]
medicine/P1	locked_full	4770/98	0.940944 [0.920427, 0.960663]	0.467949 [0.369157, 0.571382]
medicine/P1	vital_only	4770/98	0.903726 [0.881121, 0.923654]	0.213797 [0.152156, 0.294704]
medicine/P2	locked_full	4770/568	0.884963 [0.871244, 0.898355]	0.576214 [0.53961, 0.613641]
medicine/P2	vital_only	4770/568	0.777426 [0.755576, 0.797324]	0.387507 [0.352449, 0.426862]
medicine/P3	locked_full	4770/253	0.89689 [0.877769, 0.914638]	0.369064 [0.31969, 0.43299]
medicine/P3	vital_only	4770/253	0.825892 [0.799484, 0.851129]	0.245659 [0.205836, 0.296965]
medicine/S1	locked_full	4770/174	0.915409 [0.893138, 0.934396]	0.461195 [0.386994, 0.535992]
medicine/S1	vital_only	4770/174	0.836278 [0.805152, 0.867452]	0.205817 [0.161886, 0.268339]
medicine/S2	locked_full	4770/267	0.901423 [0.881709, 0.918685]	0.47988 [0.423118, 0.542435]
medicine/S2	vital_only	4770/267	0.803553 [0.77576, 0.829721]	0.241257 [0.201731, 0.297412]
medicine/S3	locked_full	4770/1606	0.763924 [0.74821, 0.778201]	0.620653 [0.595777, 0.645267]
medicine/S3	vital_only	4770/1606	0.647183 [0.62999, 0.663593]	0.502431 [0.479227, 0.526165]
medicine/S4	locked_full	4770/615	0.815018 [0.798026, 0.8323]	0.429948 [0.394407, 0.468807]
medicine/S4	vital_only	4770/615	0.700404 [0.675473, 0.723892]	0.294344 [0.264485, 0.330587]
medicine/S5	locked_full	4770/90	0.795413 [0.748333, 0.838282]	0.0797858 [0.0559498, 0.134629]
medicine/S5	vital_only	4770/90	0.720722 [0.666459, 0.771289]	0.060354 [0.0401786, 0.105613]
medicine/S6	locked_full	4770/288	0.759241 [0.732333, 0.784915]	0.167808 [0.140977, 0.20436]
medicine/S6	vital_only	4770/288	0.637132 [0.601661, 0.671803]	0.104779 [0.0908015, 0.127908]
surgery/P1	locked_full	2712/9	0.96658 [0.908112, 0.997408]	0.261247 [0.143121, 0.575463]
surgery/P1	vital_only	2712/9	0.900399 [0.726118, 0.992513]	0.232033 [0.0784594, 0.565159]
surgery/P2	locked_full	2712/41	0.920652 [0.877361, 0.956573]	0.369779 [0.236885, 0.527355]
surgery/P2	vital_only	2712/41	0.816589 [0.724202, 0.896909]	0.259537 [0.146977, 0.407764]
surgery/P3	locked_full	2712/32	0.945021 [0.887101, 0.985259]	0.582494 [0.422066, 0.800266]
surgery/P3	vital_only	2712/32	0.909142 [0.842404, 0.967197]	0.550411 [0.397203, 0.71917]
surgery/S1	locked_full	2712/17	0.962872 [0.929711, 0.988132]	0.414868 [0.217735, 0.651498]
surgery/S1	vital_only	2712/17	0.86583 [0.736999, 0.965623]	0.358354 [0.169933, 0.575844]
surgery/S2	locked_full	2712/24	0.944568 [0.907965, 0.974505]	0.425205 [0.244714, 0.626546]

Group/outcome	Arm	N/events	AUROC [95% CI]	AUPRC [95% CI]
surgery/S2	vital_only	2712/24	0.856848 [0.754038, 0.943268]	0.306171 [0.148105, 0.498264]
surgery/S3	locked_full	2712/521	0.752583 [0.728931, 0.774047]	0.429943 [0.391587, 0.470177]
surgery/S3	vital_only	2712/521	0.569154 [0.542693, 0.597046]	0.243408 [0.223789, 0.272356]
surgery/S4	locked_full	2712/140	0.860761 [0.829043, 0.889779]	0.393825 [0.319503, 0.481202]
surgery/S4	vital_only	2712/140	0.723692 [0.674209, 0.770571]	0.267537 [0.205152, 0.354107]
surgery/S5	locked_full	2712/14	0.859287 [0.794972, 0.918916]	0.0475746 [0.0183505, 0.158259]
surgery/S5	vital_only	2712/14	0.42186 [0.228028, 0.633323]	0.0228287 [0.00408892, 0.118223]
surgery/S6	locked_full	2712/42	0.856528 [0.804332, 0.902832]	0.111085 [0.0711517, 0.207811]
surgery/S6	vital_only	2712/42	0.704432 [0.612669, 0.789942]	0.0725715 [0.0375826, 0.168381]

Appendix: source-bound participant flow

The rows below are copied from source manifests and current specialty aggregate cohort summaries. Removal counts are not retained population sizes. Current Overall model-specific denominators remain a separate branch: the 7,289-record final Overall cohort is not equated with the native 7,504-record Stage1a validation population or with the specialty branches.

Scope / split	Recorded step	Records	Patients
Source inventory / all years, metadata only	sealed raw source rows (inventory only)	360,564	not recorded
Source inventory / through 2024	development source rows	287,646	not recorded
Stage1a / through 2024	eligible adult EMT rows before deduplication	34,428	not recorded
Stage1a / through 2024	duplicate rows removed	179	not recorded
Stage1a / 2020_2023	development rows before patient purge	26,745	not recorded
Stage1a / 2020_2023	development rows removed by patient isolation	1,961	not recorded
Stage1a / 2020-2023	retained	24,784	19,944
Stage1a / 2024	retained	7,504	6,600
medicine / combined through 2024	exact OP_NAME branch retained	20,765	16,274
medicine / 2020-2023	exact OP_NAME branch retained	15,995	12,203
medicine / 2024	exact OP_NAME branch retained	4,770	4,071
surgery / combined through 2024	exact OP_NAME branch retained	11,443	11,090
surgery / 2020-2023	exact OP_NAME branch retained	8,731	8,433
surgery / 2024	exact OP_NAME branch retained	2,712	2,657

The 360,564 raw-source count is seal-manifest metadata only; no raw2025 data were reopened. Original linkage quality, accrual/follow-up practice and unavailable source exclusions remain custodian facts. The flow does not infer patient identities from subtraction of unlike analysis cohorts.

Appendix: locked specialty research representatives

Copied from the accepted research-decision registry, not selected using frozen results. OOF means development out-of-fold prediction; 2024 validation/crossfit labels are retained when specified by the later source contract. The sensitivity/specificity columns reuse stored frozen aggregates at the threshold chosen before 2025. They are not new clinical evaluations.

Group / outcome	Family	Selection split	Threshold	Sensitivity	Specificity
medicine/P1	tabicl	oof_2020_2023	0.0134	0.8434	0.8779
medicine/P2	tabicl	oof_2020_2023	0.1395	0.7798	0.7993
medicine/P3	stacking	oof_2020_2023	0.4451	0.7346	0.8722
medicine/S1	catboost	validation_2024	0.0335	0.7888	0.8547
medicine/S2	tabicl	validation_2024	0.0630	0.7895	0.8406
medicine/S3	stacking	crossfit_2024	0.4962	0.7130	0.7216
medicine/S4	stacking	crossfit_2024	0.4575	0.6658	0.7813
medicine/S5	tabicl	validation_2024	0.0161	0.6723	0.7303
medicine/S6	catboost	validation_2024	0.0649	0.7000	0.6958
surgery/P1	No valid representative	none	n/a	n/a	n/a
surgery/P2	tabicl	oof_2020_2023	0.0137	0.8780	0.8177
surgery/P3	tabpfn	oof_2020_2023	0.0095	0.8205	0.9192
surgery/S1	tabicl	validation_2024	0.0042	0.9412	0.8818
surgery/S2	tabicl	validation_2024	0.0051	0.9600	0.8416
surgery/S3	stacking	crossfit_2024	0.4687	0.6292	0.7495
surgery/S4	stacking	crossfit_2024	0.4400	0.7086	0.8154
surgery/S5	stacking	crossfit_2024	0.4431	0.8000	0.6762
surgery/S6	mlp_plr	validation_2024	0.4916	0.6923	0.7245

Surgery/P1 has no valid selected single representative. The Overall accepted authority locks twelve components per outcome, not a separately recorded cross-family deployment recommendation. All 108 Overall and 216 specialty operating points are included in the CSV appendix; a new 2025 winner is not invented. Authors must confirm any existing Overall deployment decision and prospective clinical workflow.

Appendix: accepted Overall sensitivity examples

Original 2024 Overall research arms, not the different final display-model cohort. Individual-arm intervals use the exact accepted runner's outcome-stratified row bootstrap (2,000 replicates), not patient-group resampling, and remain marginal. Recorded net benefit is not transferred to a final clinical operating point when the original summary lacks a per-row threshold.

Outcome / arm	N/events	AUROC [95% CI]	AUPRC [95% CI]	Recorded NB
P1/locked_full	7504/107	0.9505 [0.9319, 0.9666]	0.4716 [0.3843, 0.5700]	0.00240
P1/vital_only	7504/107	0.8896 [0.8549, 0.9204]	0.2126 [0.1556, 0.2880]	-0.00080
P2/locked_full	7504/609	0.9138 [0.9031, 0.9240]	0.5689 [0.5325, 0.6051]	0.01333
P2/vital_only	7504/609	0.8076 [0.7888, 0.8260]	0.3411 [0.3084, 0.3802]	0.00213
P3/locked_full	7504/285	0.9193 [0.9045, 0.9338]	0.4057 [0.3558, 0.4639]	-0.03891
P3/vital_only	7504/285	0.8553 [0.8337, 0.8772]	0.2475 [0.2099, 0.2983]	-0.03931

The complete 550 Overall arm rows, 2,862 DNR comparison rows and 162 DNR flow rows accompany this report. They retain accepted age/city/CV/congestion/mortality/DNR definitions. The original fifteenth field-intubation task was not estimable; the later authorized P3 ETT-only/ETT+SGA evaluation is reported in the main text. Neither absence nor changed outcome chronology triggers a repeat fit.

Protected reproduction: seven original scientific entry points and exact local dependencies, three original runtime-lock files, an input-hash-aware plan/run route and fresh-environment tests are included. Full protected fitting and GPU-runtime reproduction were not executed; no new 2025 unsealing is dispatched by the public route.

Appendix: corrected paired text/no-text intervals

All 54 source-bound pairs and 108 split contrasts are complete. Each patient belongs once to the stratum defined by maximum outcome within the split; both probability vectors share the original seeded 2,000 patient-group draws. The overlapping-strata legacy intervals are superseded, while original models, predictions and point estimates are retained. These nominal intervals are not multiplicity-adjusted or causal effects. Earlier Overall and current specialty cohorts remain separate.

Group / outcome	Model	N / events	With-text AP	No-text AP	Delta AP [paired 95% CI]
medicine/P1	ft_transformer	4770/98	0.4125	0.4463	-0.0338 [-0.0924, 0.0249]
medicine/P1	mlp_plr	4770/98	0.4015	0.4076	-0.0061 [-0.0597, 0.0535]
medicine/P2	ft_transformer	4770/568	0.5388	0.5391	-0.0003 [-0.0113, 0.0108]
medicine/P2	mlp_plr	4770/568	0.5709	0.5654	0.0055 [-0.0032, 0.0145]
medicine/P3	ft_transformer	4770/253	0.3921	0.3042	0.0879 [0.0442, 0.1277]
medicine/P3	mlp_plr	4770/253	0.3567	0.3568	-0.0001 [-0.0111, 0.0130]
overall/P1	ft_transformer	7504/107	0.3955	0.4972	-0.1017 [-0.1632, -0.0274]
overall/P1	mlp_plr	7504/107	0.4294	0.3981	0.0312 [0.0037, 0.0580]
overall/P2	ft_transformer	7504/609	0.5439	0.5379	0.0060 [-0.0046, 0.0163]
overall/P2	mlp_plr	7504/609	0.5730	0.5667	0.0063 [-0.0027, 0.0159]
overall/P3	ft_transformer	7504/285	0.3478	0.3592	-0.0114 [-0.0345, 0.0097]
overall/P3	mlp_plr	7504/285	0.3827	0.3703	0.0124 [0.0002, 0.0264]
surgery/P1	ft_transformer	2712/9	0.1264	0.2132	-0.0868 [-0.2323, 0.0064]
surgery/P1	mlp_plr	2712/9	0.5709	0.4153	0.1557 [0.0213, 0.3525]
surgery/P2	ft_transformer	2712/41	0.4942	0.3959	0.0983 [0.0260, 0.1546]
surgery/P2	mlp_plr	2712/41	0.4240	0.4086	0.0155 [-0.0289, 0.0672]
surgery/P3	ft_transformer	2712/32	0.5219	0.5762	-0.0543 [-0.1472, 0.0792]
surgery/P3	mlp_plr	2712/32	0.6181	0.6186	-0.0004 [-0.0729, 0.0774]

AP denotes area under the precision-recall curve (AUPRC). Positive delta is with-text minus no-text. The complete nine-outcome values, OOF results, source/model identities and independent validation receipts accompany the corrected CSVs. The exact corrected statistics entrypoint is packaged for postprocessing restored saved OOF/2024 probabilities; it does not refit a model or access raw2025.